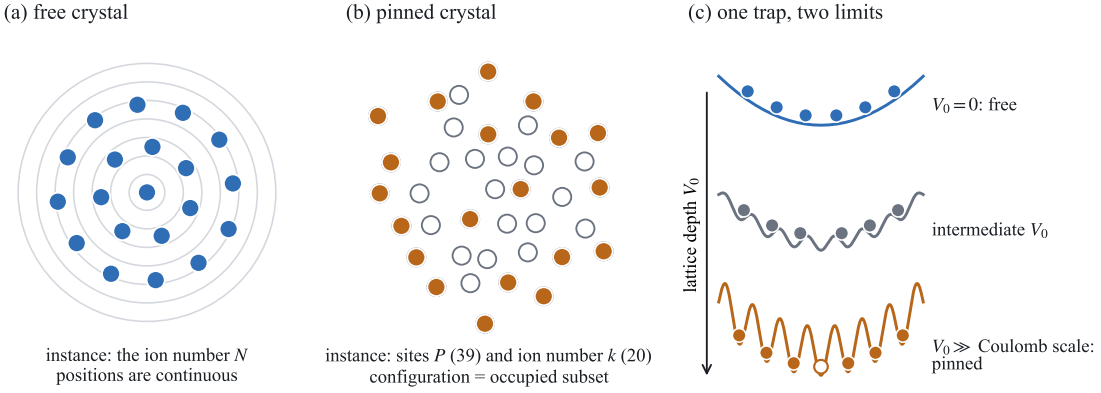
(a) free crystal
(b) pinned crystal
(c) one trap, two limits
instance: the ion number $N$
positions are continuous
instance: sites $P$ (39) and ion number $k$ (20)
configuration = occupied subset
lattice depth $V_0$
$V_0 = 0$: free
intermediate $V_0$
$V_0 \gg$ Coulomb scale:
pinned